

## **CUSTOMER CHURN PREDICTION & RETENTION ANALYTICS**

### **Data Analyst Internship Project**

**Tools Used:** Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SHAP, MySQL

**Dataset:** Telco Customer Churn Dataset

### **1. INTRODUCTION**

Customer churn is an important problem for telecommunication companies because losing existing customers can affect revenue and increase customer acquisition costs. Customer churn occurs when a customer stops using the company's services.

The objective of this project is to analyze customer churn patterns, predict customers who may be at risk of leaving, identify important factors influencing churn, and develop suitable customer retention strategies.

The project combines exploratory data analysis, machine learning, SHAP explainability, customer segmentation, and SQL analysis. Logistic Regression and Decision Tree models were developed to predict churn. SHAP was used to understand model predictions, while customer segmentation was used to classify customers into At Risk, Loyal, and Dormant groups.

#### **Objectives**

The main objectives are:

1. Clean and preprocess the customer churn dataset.
2. Analyze customer characteristics and churn patterns.
3. Identify important factors associated with churn.
4. Develop machine learning models for churn prediction.
5. Compare Logistic Regression and Decision Tree models.
6. Explain model predictions using SHAP.
7. Segment customers according to churn risk.
8. Perform SQL-based churn analysis.
9. Develop customer retention recommendations.

### **2. DATASET AND DATA PREPROCESSING**

The project uses the Telco Customer Churn dataset. The dataset contains customer demographic information, service details, contract information, payment methods, monthly charges, total charges, tenure, and churn status.

The cleaned dataset contains **7,043 customer records and 20 columns**.

Important variables include:

- Gender
- SeniorCitizen
- Partner
- Dependents
- Tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod

- MonthlyCharges
- TotalCharges
- Churn

### Data Preprocessing

The following preprocessing steps were performed:

- Removed the customer ID because it is an identifier.
- Converted TotalCharges into numerical format.
- Handled missing values in TotalCharges.
- Checked and handled duplicate records.
- Converted categorical variables using one-hot encoding.
- Converted Churn into binary values:
  - No = 0
  - Yes = 1
- Standardized numerical features for Logistic Regression.
- Divided the dataset into 80% training data and 20% testing data using stratified sampling.

### 3. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the distribution of customers and identify factors associated with churn.

The overall observed churn rate was approximately **26.54%**, while approximately **73.46%** of customers did not churn.

#### Contract Analysis

##### Contract Type    Observed Churn Rate

Month-to-month 42.64%

One year 11.27%

Two year 2.83%

Month-to-month customers showed substantially higher observed churn compared with customers on longer-term contracts.

#### Tenure Analysis

##### Tenure Group    Observed Churn Rate

0–12 Months 47.37%

13–24 Months 28.71%

25–48 Months 20.39%

49–72 Months 9.51%

The analysis indicates that customers with shorter tenure had higher observed churn.

#### Internet Service Analysis

##### Internet Service    Observed Churn Rate

DSL 18.89%

Fiber optic 41.78%

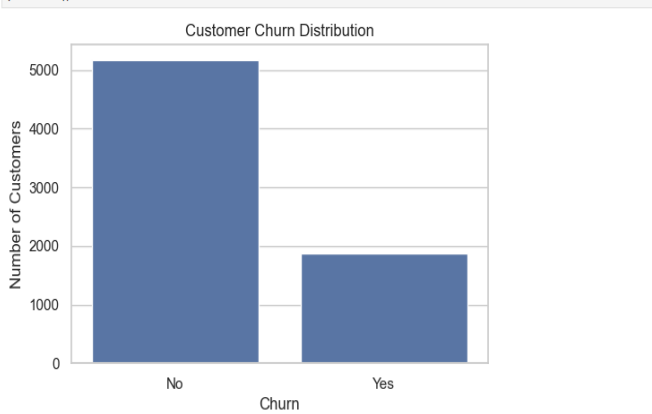
No internet 7.21%

Payment Method Analysis

Payment Method Observed Churn Rate

|                  |        |
|------------------|--------|
| Electronic check | 45.15% |
| Mailed check     | 18.92% |
| Bank transfer    | 16.71% |
| Credit card      | 15.24% |

These analyses helped identify customer groups that require further investigation and retention planning.



4. MACHINE LEARNING AND CUSTOMER SEGMENTATION

4.1 Churn Prediction

Two machine learning classification models were developed:

- 1. Logistic Regression
- 2. Decision Tree

The models were trained using the processed customer data and evaluated using accuracy, precision, recall, and F1-score.

Logistic Regression Results

Metric Result

|           |        |
|-----------|--------|
| Accuracy  | 79.84% |
| Precision | 65.00% |
| Recall    | 52.14% |
| F1 Score  | 57.86% |

```
print("F1 Score :", round(logistic_f1, 4))

Logistic Regression Performance
-----
Accuracy : 0.7984
Precision: 0.65
Recall   : 0.5214
F1 Score : 0.5786
```

The Logistic Regression model provides a probability-based estimate of customer churn.

Decision Tree

A Decision Tree classifier with a maximum depth of 5 was developed. The model was evaluated using the same classification metrics and its performance was compared with Logistic Regression.

```
model_comparison.round(4)

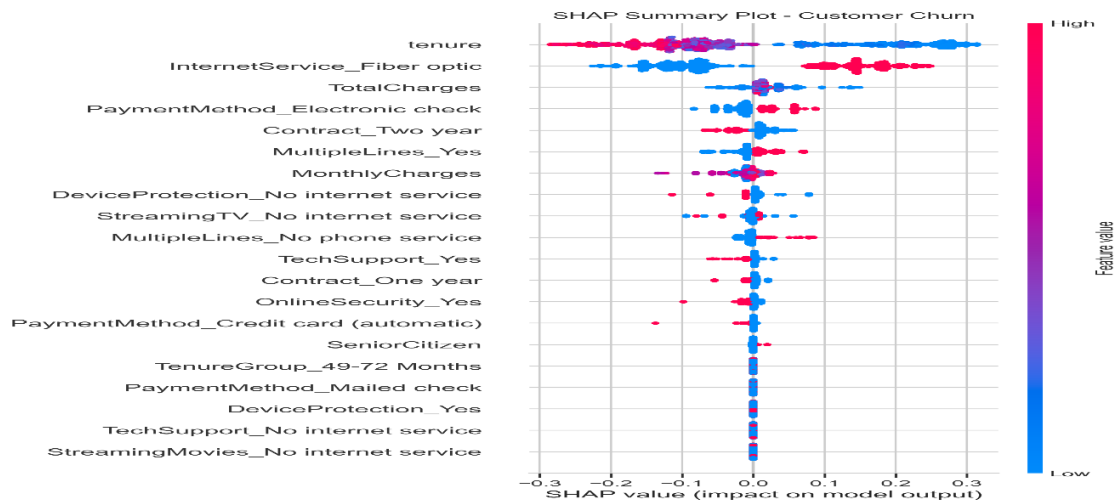
76]:
   Model  Accuracy  Precision  Recall  F1 Score
0  Logistic Regression  0.7984  0.6500  0.5214  0.5786
1    Decision Tree      0.7942  0.6312  0.5401  0.5821

77]: comparison_path = r"F:\Data Analyst Internship\Customer Churn Prediction & Retention Analyt
```

4.2 SHAP Explainability

SHAP (SHapley Additive exPlanations) was used to explain the Decision Tree predictions. SHAP helps identify how different customer features contribute to the model's prediction.

This makes the machine learning model easier to interpret and helps connect the prediction results with business analysis.



### 4.3 Customer Segmentation

Customers were divided into three project-defined analytical segments using predicted churn probability and tenure.

#### Segment Number of Customers

Loyal 737

Dormant 372

At Risk 300

**At Risk:** Customers with predicted churn probability of at least 50%.

**Loyal:** Customers with at least 24 months of tenure and predicted churn probability below 50%.

**Dormant:** Customers who do not fall into the At Risk or Loyal definitions.

The segmentation allows retention activities to be organized according to customer risk.

```
segment_analysis.round(2)
```

|   | Customer_Segment | Customers | Avg_Tenure | Avg_Monthly_Charges | Avg_Churn_Probability | Customer_Percentage |
|---|------------------|-----------|------------|---------------------|-----------------------|---------------------|
| 0 | At Risk          | 300       | 9.89       | 79.21               | 0.66                  | 21.29               |
| 1 | Dormant          | 372       | 10.12      | 45.22               | 0.25                  | 26.40               |
| 2 | Loyal            | 737       | 51.88      | 67.46               | 0.12                  | 52.31               |

### 5. SQL ANALYSIS, FINDINGS AND RECOMMENDATIONS

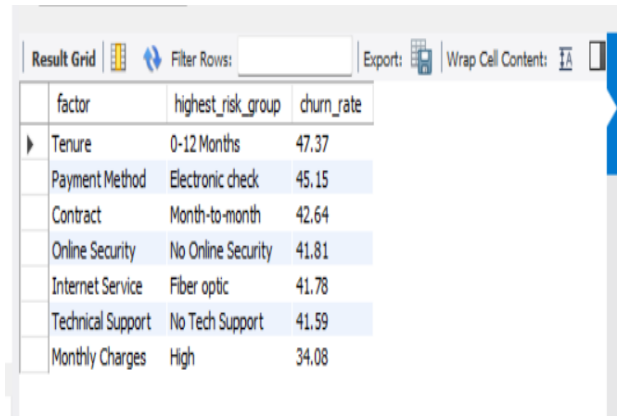
MySQL was used to perform additional structured churn analysis. The SQL analysis covered overall churn, contract type, tenure, internet service, payment method, monthly charges, technical support, and online security.

#### Overall Churn

| Metric                | Value  |
|-----------------------|--------|
| Total Customers       | 7,043  |
| Churned Customers     | 1,869  |
| Non-Churned Customers | 5,174  |
| Overall Churn Rate    | 26.54% |

### Major Churn Factors

| Factor            | Higher-Risk Group  | Churn Rate |
|-------------------|--------------------|------------|
| Contract          | Month-to-month     | 42.64%     |
| Tenure            | 0–12 Months        | 47.37%     |
| Internet Service  | Fiber optic        | 41.78%     |
| Payment Method    | Electronic check   | 45.15%     |
| Monthly Charges   | High               | 34.08%     |
| Technical Support | No Tech Support    | 41.59%     |
| Online Security   | No Online Security | 41.81%     |



| factor            | highest_risk_group | churn_rate |
|-------------------|--------------------|------------|
| Tenure            | 0-12 Months        | 47.37      |
| Payment Method    | Electronic check   | 45.15      |
| Contract          | Month-to-month     | 42.64      |
| Online Security   | No Online Security | 41.81      |
| Internet Service  | Fiber optic        | 41.78      |
| Technical Support | No Tech Support    | 41.59      |
| Monthly Charges   | High               | 34.08      |

### Retention Recommendations

Based on the analysis, the following actions are recommended:

1. **Focus on At Risk customers:** Prioritize customers with higher predicted churn probability for targeted retention campaigns.
2. **Improve early onboarding:** Customers in the first year showed higher observed churn, so stronger onboarding and early support can be provided.
3. **Encourage suitable long-term contracts:** Month-to-month customers showed higher observed churn, so suitable longer-term plans can be promoted.
4. **Improve technical support:** Customers without technical support showed higher observed churn, indicating an opportunity to improve awareness and access to support.
5. **Review high monthly charges:** High-charge customers can be provided with suitable plans, bundles, or pricing options.
6. **Monitor high-risk groups:** Customer groups with higher observed churn should be regularly monitored for service and customer-experience issues.

### Conclusion

This project developed a complete customer churn analytics workflow using Python, machine learning, SHAP, customer segmentation, and SQL.

The analysis identified differences in observed churn across contract type, tenure, internet service, payment method, monthly charges, and support services. Machine learning models were used to estimate churn risk, while SHAP provided explainability for model predictions. Customer segmentation further classified customers into At Risk, Loyal, and Dormant groups.

The results provide a data-driven framework for identifying customers who may require additional retention attention and for developing targeted customer retention strategies.

### Future Scope

The project can be extended by testing additional machine learning algorithms such as Random Forest and XGBoost, performing hyperparameter tuning, incorporating customer complaints and service interactions, developing real-time churn prediction, and creating an automated dashboard for continuous churn monitoring.

### References

1. Telco Customer Churn Dataset, Kaggle.
2. Python Documentation.
3. Pandas Documentation.
4. Scikit-learn Documentation.
5. SHAP Documentation.
6. MySQL Documentation.